Ex. No: 05	IMPLEMENTATION OF NAVIGATION AND ROUTING
Date of Exercise	08.09.2026
Git-hub Host Link	

Aim:

To implement navigation between multiple Flutter screens using Navigator and named routes.

Description:

This experiment demonstrates how to navigate between multiple screens in a Flutter application. The application contains **Home, Profile, and Details** screens. Named routes are used for navigation, and Navigator.pop() is used to return to the previous screen. A colorful **purple, pink, and sky-blue** theme is used to improve the user interface.

Procedure:

Step 1: Create a new Flutter project using VS Code or the terminal.

Step 2: Create separate Dart files for the Home, Profile, and Details screens inside the lib folder.

Step 3: Design the Home screen with a button to navigate to the Profile screen.

Step 4: Design the Profile screen with a button to navigate to the Details screen.

Step 5: Design the Details screen with a Back button to return to the previous screen.

Step 6: Define named routes for Home, Profile, and Details inside MaterialApp.

Step 7: Use Navigator.pushNamed() to navigate between the screens.

Step 8: Use Navigator.pop() to return to the previous screen.

Step 9: Apply a colorful user interface using purple, pink, and sky-blue colors, gradients, cards, icons, and buttons.

Step 10: Run the application using flutter run and check the navigation between all screens.

Step 11: Verify the navigation flow: **Home → Profile → Details** and back to the previous screens.

Step 12: Confirm that navigation and routing work correctly in the Flutter application.

Program:

main.dart -

```
import 'package:flutter/material.dart';
import 'home.dart';
import 'profile.dart';
import 'details.dart';

void main() {
  runApp(const NavigationApp());
}

class NavigationApp extends StatelessWidget
  { const NavigationApp({super.key});

  @override
  Widget build(BuildContext context)
    { return
    MaterialApp( debugShowCheckedMode
    Banner: false, title: 'Navigation App',

      theme: ThemeData(
        colorScheme:
          ColorScheme.fromSeed( seedColor:
          Colors.purple,
        ),
        useMaterial3: true,
      ),

      initialRoute: '/',

      routes: {
        '/': (context) => const HomePage(),
        '/profile': (context) => const ProfilePage(),
        '/details': (context) => const DetailsPage(),
      },
    );
  }
}
```

home.dart -

```
import 'package:flutter/material.dart';
class HomePage extends StatelessWidget
{ const HomePage({super.key});
 @override
 Widget build(BuildContext context)
  { return Scaffold(
   appBar:
    AppBar(     title:
    const
    Text('Home',
     style:
      TextStyle( color:
      Colors.white,
      fontWeight: FontWeight.bold,
     ),
    ),
    centerTitle: true,
    backgroundColor: Colors.purple,
   ),
   body: Container(
    decoration: const
    BoxDecoration( gradient:
    LinearGradient( colors: [
        Colors.purple,
        Colors.pink,
        Colors.lightBlue,
       ],
       begin: Alignment.topLeft,
       end: Alignment.bottomRight,
     ),
    ),
    child: Center(
     child: Card(elevation: 10,
      margin: const EdgeInsets.all(30),
      shape: RoundedRectangleBorder(
       borderRadius: BorderRadius.circular(25),
      ),
      child: Padding(
       padding: const EdgeInsets.all(35),
       child: Column(
        mainAxisSize: MainAxisSize.min,
        children: [
         const
          Icon( Icons.home_round
          ed, size: 80,
```

```
  color: Colors.purple,
),
const SizedBox(height: 20),
const Text(
  'Welcome Home!',
  style:
  TextStyle( fontSiz
  e: 28,
    fontWeight: FontWeight.bold,
    color: Colors.purple,
  ),
),
const SizedBox(height: 10),
const Text(
  'Explore the different screens',
  style: TextStyle(
    fontSize: 16,
    color: Colors.grey,
  ),
),
const SizedBox(height: 30),
ElevatedButton.icon( onPresse
d: () {
    Navigator.pushNamed(context,  '/profile');
  },
  icon: const Icon(Icons.person),
  label: const Text('Go to Profile'),
  style:
  ElevatedButton.styleFrom( back
  groundColor: Colors.pink,
  foregroundColor: Colors.white,
  padding: const
      EdgeInsets.symmetric( horizontal: 25,
      vertical: 15,
    ),
    shape:
      RoundedRectangleBorder( borderRadius:
      BorderRadius.circular(30),
       ),),),],),),),),),);}}
```

profile.dart -

```
import 'package:flutter/material.dart';
class ProfilePage extends StatelessWidget
  { const ProfilePage({super.key});
  @override
  Widget build(BuildContext context)
    { return Scaffold(
      appBar:
        AppBar(      title:
        const
        Text( 'Profile',
          style:
            TextStyle( color:
            Colors.white,
            fontWeight: FontWeight.bold,
          ),
        ),
        centerTitle: true,
        backgroundColor: Colors.pink,
      ),
      body: Container(
        decoration: const
        BoxDecoration( gradient:
        LinearGradient(  colors: [
              Colors.pink,
              Colors.purple,
              Colors.lightBlue,
            ],
            begin: Alignment.topRight,
            end: Alignment.bottomLeft,
          ),
        ),
        child:
          Center( child:
          Card( elevation:
          10,
            margin: const EdgeInsets.all(30),
            shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(25),
            ),
            child: Padding(
              padding: const EdgeInsets.all(35),
              child: Column(
                mainAxisSize: MainAxisSize.min,
                children: [
                  const
```

```
CircleAvatar( radiu
  s: 50,
  backgroundColor: Colors.pink,
  child: Icon(Icons.person,

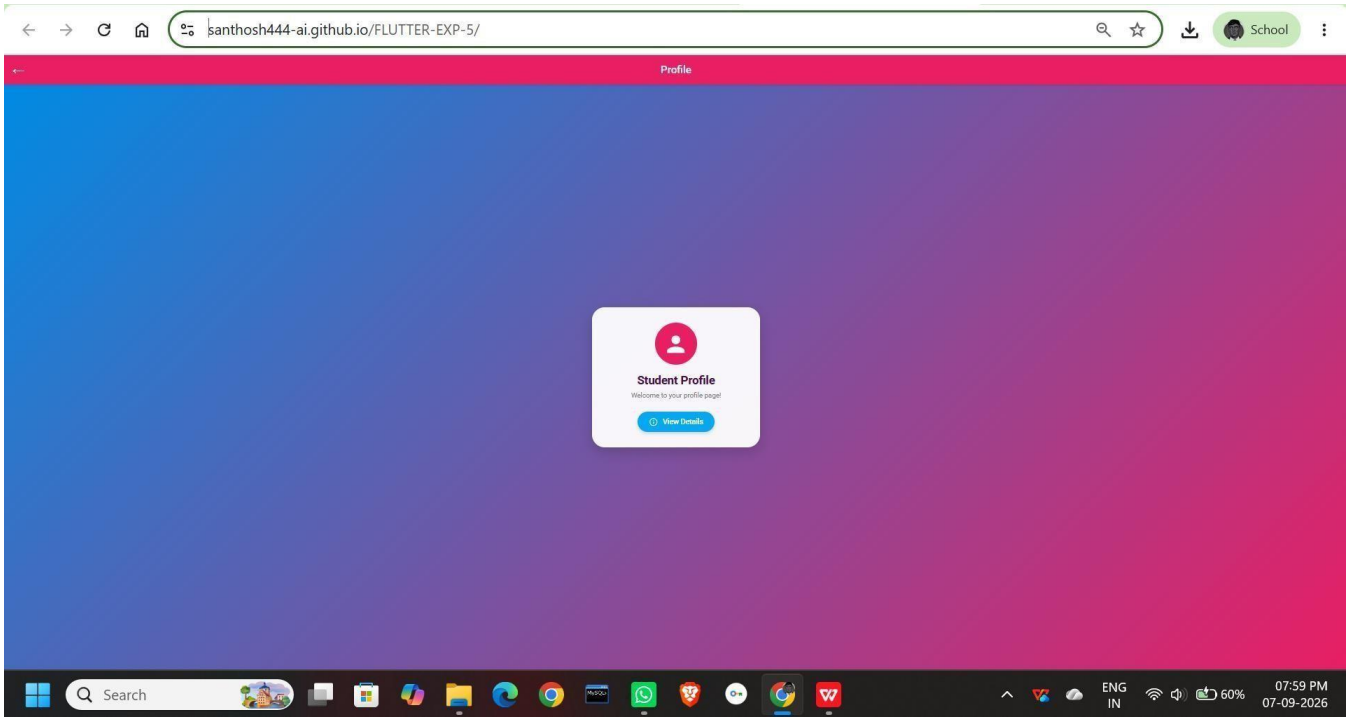
    size: 60,
    color: Colors.white,
  ),
),
const SizedBox(height: 20),
const Text('Student Profile',
style: TextStyle(
    fontSize: 27,
    fontWeight: FontWeight.bold,
    color: Colors.purple,
  ),
),
const SizedBox(height: 10),
const Text('Welcome to your profile page!',
  style: TextStyle(
    fontSize: 16,
    color: Colors.grey,
  ),
),
const SizedBox(height: 30),
ElevatedButton.icon( onPresse
d: () {
    Navigator.pushNamed(context, '/details');
  },
  icon: const Icon(Icons.info_outline),
  label: const Text('View Details'),
  style:
  ElevatedButton.styleFrom( backgrou
  ndColor: Colors.lightBlue,
  foregroundColor: Colors.white,
  padding: const
      EdgeInsets.symmetric( horizontal: 25,
      vertical: 15,
    ),
    shape:
      RoundedRectangleBorder( borderRadius:
      BorderRadius.circular(30),
       ),),),],),),),),),);}
```

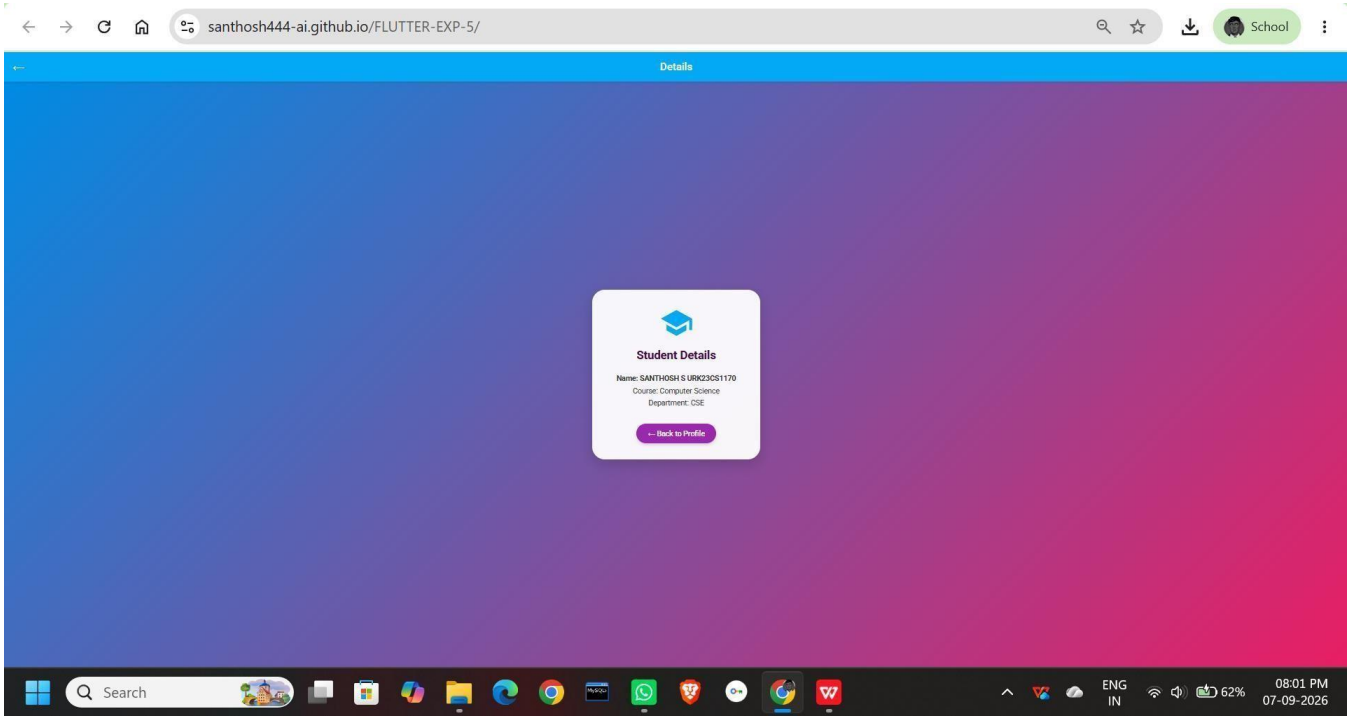
details.dart -

```
import 'package:flutter/material.dart';
class DetailsPage extends StatelessWidget
  { const DetailsPage({super.key});
  @override
  Widget build(BuildContext context)
    { return Scaffold(
     appBar:
       AppBar(     title:
       const
       Text( 'Details',
         style:
           TextStyle( color:
           Colors.white,
           fontWeight: FontWeight.bold,
         ),
       ),
       centerTitle: true,
       backgroundColor: Colors.lightBlue,
     ),
     body: Container(
       decoration: const
       BoxDecoration( gradient:
       LinearGradient(  colors: [
            Colors.lightBlue,
            Colors.purple,
            Colors.pink,
          ],
          begin: Alignment.topLeft,
          end: Alignment.bottomRight,
        ),
       ),
       child:
         Center( child:
         Card( elevation:
         10,
           margin: const EdgeInsets.all(30),
           shape: RoundedRectangleBorder(
             borderRadius: BorderRadius.circular(25),
           ),
           child: Padding(
             padding: const EdgeInsets.all(35),
             child: Column(mainAxisSize: MainAxisSize.min,
               children: [
                 const
                   Icon( Icons.school_roun
```

```
ded, size: 80,
  color: Colors.lightBlue,
),
const SizedBox(height: 20),
const Text('Student Details',
style: TextStyle(
    fontSize: 27,
    fontWeight: FontWeight.bold,
    color: Colors.purple,
  ),
),
const SizedBox(height: 20),
const Text(
  'Name: SANTHOSH S URK23CS1170\n'
  'Course: Computer Science\n'
  'Department: CSE',
  textAlign: TextAlign.center,
  style: TextStyle(
    fontSize: 17,
    height: 1.7,
    color: Colors.black87,
  ),
),
const SizedBox(height: 30),
ElevatedButton.icon(onPressed: () {
    Navigator.pop(context);
  },
  icon: const Icon(Icons.arrow_back),
  label: const Text('Back to Profile'),
  style:
  ElevatedButton.styleFrom( background
  Color: Colors.purple, foregroundColor:
  Colors.white, padding: const
  EdgeInsets.symmetric(
      horizontal: 25,
      vertical: 15,
    ),
    shape:
      RoundedRectangleBorder( borderRadius:
      BorderRadius.circular(30),
    ),),),],),),),),),);}}
```

Output:


santhosh444-ai.github.io/FLUTTER-EXP-5/
School
Profile
Student Profile
Welcome to your profile page!
View Details
Search
ENG
IN
60%
07:59 PM
07-09-2026


santhosh444-ai.github.io/FLUTTER-EXP-5/
School
Details
Student Details
Name: SANTHOSH S URK23CS1170
Course: Computer Science
Department: CSE
← Back to Profile
Search
ENG
IN
62%
08:01 PM
07-09-2026

Result:

The Flutter application was successfully implemented with navigation between Home, Profile, and Details screens using named routes and back navigation.